Linear Model :-

→ Describes the relationship b/w a dependent variable and one or more independent variables

→ Assumes a linear relationship, where the dependent variable is a linear combination of independent variables

Goal : To find the best-fitting line or hyperplane that minimizes the error between predicted and actual values.

Examples of Linear Models:

1) Simple Linear Regression
2) Multiple Linear Regression
3) Ridge and Lasso Regression : Extension of linear regression that add regularization to prevent overfitting

Non-Linear Model

→ does not assume a linear relationship

→ It can capture more complex interactions between variables

→ Allows the use of curves, exponentials, logarithms in data modelling

Examples of Non-Linear Models:

1. Logistic Regression
2. Neural Networks : Inspired by biological neurons, they can learn from large datasets and capture complex relationships.

→ Non-Linear Models offer more flexibility but are harder to interpret and estimate compared to linear models.

→ Linear Models use methods like Ordinary Least Squares (OLS),

→ Non-Linear models rely on advanced techniques like optimization algorithms

Multi class classification :-

→ An instance can belong to only one class from a set of more than 2 mutually exclusive classes.

→ The model predicts a single label for each instance from multiple possible categories

Examples:-

i) Handwritten digit Recognition : classify digits into one of 10 classes (0-9)

ii) Sentiment Analysis : classify text as +ve, -ve or neutral

iii) Disease diagnosis : classify a medical condition as diabetes, heart disease or healthy

Multi-label classification :-

→ An instance can be assigned to multiple classes simultaneously, meaning, the classes are not mutually exclusive.

→ The model predicts multiple labels for each instance

Examples:-

i) Image Tagging : An image may contain multiple objects [e.g 'cat, dog, tree.

ii) Text categorization : A document may belong to multiple categories

iii) Medical diagnosis : A patient may be diagnosed with multiple conditions, like hypertention and diabetes.

Challenges:-

Multi class classification :- 1) requires a clear distinction between classes which can be challenging if some classes are ambiguous.

2) Performance may suffer if there is an imbalanced dataset.

Multi label classification : 1) The model must predict a combination of labels making it more complex.

2) Needs to handle correlations b/w labels [Ex:- dog & animal often co-occur]

∴ Both are essential techniques, and the choice b/w them depends on the specific task and data structure in use

Regression :

→ It is used to model the relationship b/w a dependent variable and one or more independent variables (predictors).

→ It predicts continuous values like temperature, salary, price etc by estimating how the dependent variable changes when independent variables are adjusted

Key Concepts

1. Dependent / target variable : The main factor we want to predict
2. Independent variable : Variables that help predict the target var
3. Outliers : Extreme values that may distort the model's accuracy
4. Multicollinearity : A situation where independent variables are highly correlated with each other, which can affect the accuracy of ranking the impact of diff variables
5. Underfitting : The model performs poorly even on training data
6. overfitting : model performs well on training data but fails on unseen test data

Uses of regression :

1) Prediction : predicting real-world values like sales, weather etc
2) Analyzing how variables change over time
3) Identifying and quantifying the relationships b/w variables

→ LR allows to predict outcomes based on new input data

→ The goal is to find the best fit linear relationship b/w these variables to make predictions

Assumptions

→ relationship b/w dependent & independent variables should be linear

→ No multicollinearity in the data.

→ Multivariate Normality

Simple Linear Regression :-

→ It models the relationship b/w a dependant variable and a single independent variable.

→ The relationship is represented by a straight line

→ The key aspects of simple Linear Regression are:

i) the dependent variable must be continuous

ii) the independ variable can be continuous or categorical

The Goal is to predict the dependent variable using this single input.

Equation: $\boxed{Y = aX + b}$

Y - target variable X - predictor variable

a = slope [how much Y changes for a unit change in X)

b = intercept

→ Example:

Multiple Linear Regression :

→ Involves more than one independent variable.

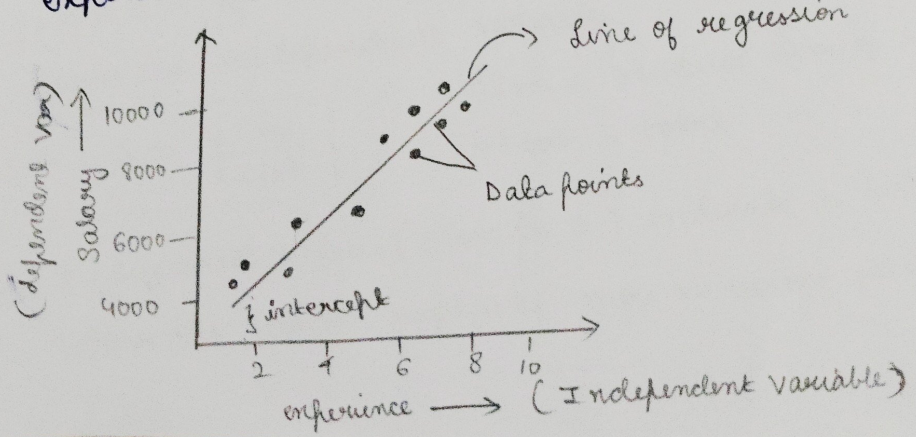
→ The goal is to predict the dependent variable based on multiple inputs [independent variables]

Eq:- $\boxed{Y = b_0 + b_1X_1 + b_2X_2 + \cdots b_nX_n}$

b_0 - intercept

$b_1, b_2, \cdots b_n$ are the coefficients (slopes) of independent variables $X_1, X_2, \cdots X_n$

EX:- Predicting an Employee's salary based on their years of experience [independent variable]



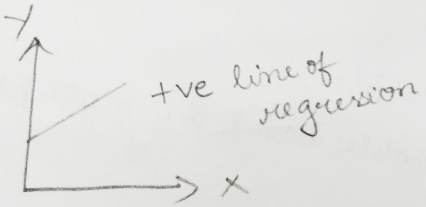
Mathematical approach:

1. Residual / Error = Actual value − Predicted value
2. Sum of residuals / Errors = sum of differences b/w actual and predicted values.
3. Best fit line : The line that minimizes the sum of squared Errors (SSE), ensuring the smallest difference b/w the actual values and the predicted values on the line

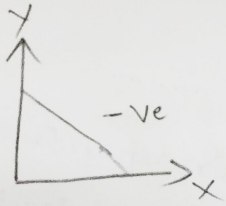
Types of Linear relationships :-

+ve linear relationship :- When both dependent and independent variables increase together

−ve linear relationship : When the dependent variable decreases as the independent variable increases



$Y = a_0 + a_1 X$



$Y = -a_0 + a_1 X$

→ The best fit line is the one that minimizes the error b/w the predicted values (on the regression line) and the actual observed values.

→ This is done by minimizing the sum of squared errors

→ Least error leads to the most accurate model

Model Evaluation Metrics :-

1. Mean Absolute Error (MAE):-

→ Measures the average absolute diff b/w actual & predicted values.

$$MAE = \frac{1}{n} \sum_{i=1}^{n} |y_i - \hat{y}_i|$$ (less sensitive to outliers)

→ Lower the MAE, the closer the predictions are to actual values, indicating a better fit

2. Mean Squared Error (MSE):-

→ calculates the average of the squared differences b/w actual & predicted values, giving more weight to larger errors.

$$MSE = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$ (sensitive to outliers)

→ Smaller MSE indicates better accuracy, as it suggests predictions are closer to the actual values.

3. Root Mean Squared Error (RMSE):-

RMSE is the square root of MSE and provides error values in the same units as the target variable, making it more interpretable

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2}$$

→ Like MSE, a smaller RMSE indicates a model that fits the data better

→ RMSE is sensitive to outliers due to squared term.

4. R-squared :- (R^2):

→ R^2, also known as coefficient of determination, measures the proportion of variance in the dependent variable that is predictable from the independent variables

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

SS_{res} - residual sum of squares

SS_{tot} - total " " "

Higher R-squared values (close to 1) indicate a better model fit, where the model explains more variance in the data

Classification :-

→ A supervised learning technique used to categorize new observations based on training data.

→ classification predicts categorical outputs like Yes/No cat/Dog . Spam/Not Spam.

Types of classifications :-

1. Binary :- 2 possible outcomes [E.g Male/female, Yes/No]
2. Multi-class :- More than 2 outcomes [Eg: Music genres]

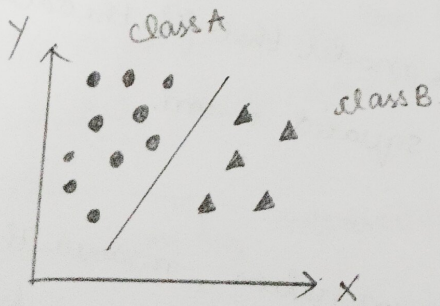
Learners in classification :-

1. Lazy learners : Store training data and classify based on the most related stored data when testing [Predict later]

   Ex :- KNN, etc.

2. Eager learners : Build a classification model during training and use it to make predictions [predict immediately after training]

   Ex :- Decision Trees, Naïve Bayes etc



Linear Models
- Logistic regression
- SVM

Non-linear Models
- KNN
- Naive Bayes
- decision trees
- random forest

Bayes Theorem

It calculates the conditional probability of an event given prior info.

$$P(Y/X) = \frac{P(X/Y)\ P(Y)}{P(X)}$$

$P(Y/X)$: posterior probability of class Y given attribute X

$P(X/Y)$: likelihood of observing X given Y

$P(Y)$: prior probability of class Y

$P(X)$: overall probability of X

Naive Bayes classifier :-

→ It is a simple probabilistic classifier based on Bayes' Theorem

→ It assumes that the presence of a particular feature in a class is independent of any other feature

→ It works well in real world situations, especially for text classification spam filtering etc.

→ The key idea behind the Naive Bayes classifier is to simplify the calculation of probabilities by assuming that all features $X_1, X_2, \dots X_n$ are conditionally independent of each other, given the class label Y.

→ Thus, instead of calculating joint probability $P(X_1, X_2, \dots X_n/Y)$ which can be expensive, we calculate

$$P(X/Y) = P(X_1/Y) \cdot P(X_2/Y) \cdot P(X_3/Y) \dots P(X_n/Y)$$

It reduces the computational cost & simplifies the model.

How a Naive Bayes classifier Works?

→ The Naive Bayes classifier operates by applying Bayes Theorem to classify new instances based on their feature values.

→ The idea is to compute the posterior probability of a class Y, given the values of the feature set $x = (x_1, x_2, \cdots, x_n)$

$$P(Y/X) = \frac{P(X/Y) \cdot P(Y)}{P(X)}$$

∴ $P(X)$ is constant for all classes, the Naive Bayes classifier simplifies the computation to maximize the product of prior probability $P(Y)$ and the likelihood $P(X/Y)$:

$$\hat{Y} = \arg\max P(Y) \cdot \prod_{i=1}^{n} P(x_i/Y)$$

EX:① - Naive Bayes classifier

	Day	Outlook	Temp	Humidity	Wind	Play Tennis (Target Variable)

New Instance → (Outlook = Sunny, Temp = Cool, Humidity = High, Wind = Strong)

Check whether the New Instance can be classified as Yes or No

Sol'n:- P(PlayTennis = yes) = 9/14 = 0.64
P(PlayTennis = no) = 5/14 = 0.36
} Prior probabilities

Conditional probabilities of individual attributes:-

Outlook	Y	N
Sunny	2/9	3/5
overcast	4/9	0
rain	3/9	2/5

Humidity	Y	N
high	3/9	4/5
normal	6/9	1/5

Temperature	Y	N
hot	2/9	2/5
mild	4/9	2/5
cool	3/9	1/5

Windy	Y	N
Strong	3/9	3/5
weak	6/9	2/5

$$v_{NB} = \underset{v_j \in \{yes, no\}}{\operatorname{argmax}} P(v_j) \prod_i P(a_i/v_j)$$

$$= \underset{v_j \in \{yes, no\}}{\operatorname{argmax}} P(v_j) \; P(Outlook = sunny/v_j) \cdot P(Temp = cool/v_j) \cdot P(Humidity = high/v_j) \cdot P(Wind = strong/v_j)$$

$$v_{NB}(yes) = P(yes) \; P(sunny/yes) \; P(cool/yes) \; P(high/yes) \; P(strong/yes)$$

$$= 9/14 \quad 2/9 \quad 3/9 \quad 3/9 \quad 3/9$$

$$= 0.0053$$

$$v_{NB}(NO) = P(no) \; P(sunny/no) \; P(cool/no) \; P(high/no) \; P(strong/NO)$$

$$= 0.0206$$

∴ Person is not going to play tennis